MAXIMILIANO ROJAS LEMA

Electronic Civil Engineer / Master of Science in Engineering

Profile

As an engineer and researcher, I am equipped to develop comprehensive solutions that encompass both the problem and its context. I enjoy tackling challenges in a structured way, from their definition to the possible steps and goals for execution, always keeping in mind and adapting to any unforeseen events that may arise. My foundations are resilience, creativity, curiosity, and applied knowledge.

Work Experience

present

Virtual Humans Developer

Chaturbate - Remote

- Cognitive architectures with LLM in Docker/Python, modeling in Daz Studio, UE5 for immersive spaces.
- System integration via APIs.
- Development of interactive virtual humans in Daz3D and Unreal Engine.

Robotics Engineer

Renshi Robotics Studio, Shenzhen, China - Remote

- **Semantic navigation with LLM+RL**: Framework combining semantic maps with LLM to interpret natural language instructions and guide RL agents in low-level control.
- **Biological flexible structure**: Integration of stomach model with spherical joint control ($\pm 30°$), collision detection, and joint randomization in Isaac Sim.
- **Mobile robot in Isaac Lab**: Adaptation of paper Sim-to-Real for ROS 2, RL training environments.
- **Synthetic detection**: Complete pipeline with Isaac Replicator, 6 object categories, domain randomization, and training for real-world applications.

Robotics Engineer

Private Client, USA - Remote

- Isaac Sim 4.0 simulation for startup validation: Spot robot assembles 2×4 planks with realistic physics (mass/friction/collision), collision detection, and precise placement.
- Technical analysis and considerations for robotics applications in cell cultivation and how the process can be simulated in Isaac Sim.

Robotics Engineer

Nazarbayev University Institute of Smart Systems and Artificial Intelligence - Astana, Kazakhstan - Remote

- Mentorship of a master's candidate in deep reinforcement learning applied to robotics.
- Technical problem-solving in reinforcement learning.
- Design of digital twins for simulation in Isaac Sim and Isaac Lab.
- Design and testing of training pipelines.

Contact

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Upwork Profile
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Technologies

- Python
- C++
- PyTorch
- CAD Design (Fusion 360)
- Robotics Simulation / Digital Twins (Isaac Sim)
- ROS/ROS2
- Unreal Engine
- PCB Design (Fusion 360 Circuits)

Languages

Spanish	Native
English	B2

Robotics Engineer

Summon, France - Remote

- Environment analysis for humanoid simulation and advisory in robotic design.

Robotics Assistant Developer

PUCV School of Computer Science

- Design of a humanoid robot for interaction with children with autism spectrum disorder (PUCV university research).
- Resulting in "Design Path for a Social Robot for Emotional Communication for Children with Autism Spectrum Disorder (ASD)"

Assistant Developer

Valparaíso Makerspace

- Internship
- Design of didactic introductory electronics experiences.
- Technical assistant Fondecyt (R&D): "New challenges for education in Chile: Supports for independent living of adults with intellectual or developmental disabilities" (1190789).

Education

Q1 WoS Scientific Journal Publication

MDPI - "Automation Control and Robotics in Human-Machine Cooperation"

- "Easy-to-use deep reinforcement learning library for AI-based mobile robots for Isaac Sim" Applied Sciences 12, no. 17: 8429.
- Click here to view the article.

Specialization in Modern Robotics: Mechanics, Planning and Control, Electronic Engineering, Robotics and Mechatronics

Northwestern University, Coursera

- Kinematics and Dynamics.
- Motion Planning and Control.
- Robotic Manipulation and Mobile Robots.
- Mobile Manipulation.
- Click here to view the certificate.

Foundations of Project Management

Google, Coursera

- Click here to view the certificate.

Master of Science in Engineering, Electrical Engineering

PUCV

- Thesis: "Deep reinforcement learning library for AI-based mobile robots in Isaac Sim".

Electronic Civil Engineer Degree

PUCV

- Thesis: "Creation of a service robot platform for Deep Learning and robotics applications".